

# No nonzero irreducible representation on a separable Hilbert space

MathlibAnnex.CStarAlgebra.CAR.not\_isIrreducible\_of\_separable

## theorem

Rules out irreducible representations of  $T$  on separable Hilbert spaces by the finite-dimensionality theorem for a single irreducible representation class.

## Statement

Let  $A$  be the completed CAR algebra. Let  $\pi : A \rightarrow B(\mathcal{H})$  be the direct sum of one chosen GNS representation from each pure-state GNS equivalence class, and let  $(L_i)_{i \in I}$  be the unitaries obtained from the fixed CAR homogeneity and shell construction. Put  $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ , and write  $\iota(a) = \pi(a)$  for the embedding into  $T$ . For every separable complex Hilbert space  $H$ , there is no nonzero irreducible  $*$ -representation

$$\rho : T \longrightarrow B(H).$$

Preservation of the identity is not assumed. Here irreducibility means that the only closed subspaces of  $H$  invariant under every  $\rho(t)$  are  $\{0\}$  and  $H$ ; for a  $*$ -representation these subspaces also reduce the representation.

## Assumptions

The algebra  $T$  is the particular algebra constructed above, not an arbitrary  $C^*$ -algebra. Its established properties are infinite dimensionality and a faithful irreducible inclusion  $\rho_0 : T \rightarrow B(\mathcal{H})$  to which every nonzero irreducible representation is unitarily equivalent. Separability is required of  $H$ , not of  $T$ ; faithfulness and preservation of the identity are not assumptions on  $\rho$ .

## Conclusion

Every nonzero  $*$ -representation of  $T$  on a separable Hilbert space is reducible. This does not rule out irreducible representations on nonseparable spaces: the inclusion  $\rho_0$  is one.

The theorem places no separability restriction on the Hilbert space of another irreducible representation used for comparison. It asserts nonexistence only for the proposed representation on  $H$ .

## Proof route

Suppose that a nonzero irreducible representation  $\rho$  exists on a separable  $H$ . It must preserve the identity. Composing its unitary equivalence with  $\rho_0$  with the equivalence for an arbitrary irreducible representation shows that  $\rho$  represents the unique irreducible representation class. The cited finite-dimensionality theorem then contradicts the infinite dimensionality of  $T$ .

### ▼ Proof steps

1. Put  $P = \rho(1)$ . The  $*$ -representation identities give  $P^* = P$ ,  $P^2 = P$ , and  $P\rho(t) = \rho(t)P = \rho(t)$ . Thus  $P$  is an orthogonal projection with reducing range. Since  $\rho \neq 0$ , one has  $P \neq 0$ ; irreducibility therefore forces  $P = I_H$ . The same representation  $\rho$  is consequently unital; no new map is introduced.
2. The classification theorem gives a unitary  $U_\rho : \mathcal{H} \rightarrow H$  with  $U_\rho \rho_0(t) = \rho(t)U_\rho$ . For any nonzero irreducible  $*$ -representation  $\sigma : T \rightarrow B(K)$ , the same theorem gives a unitary  $U_\sigma : \mathcal{H} \rightarrow K$  with  $U_\sigma \rho_0(t) = \sigma(t)U_\sigma$ . There is no separability assumption on  $K$ .
3. Set  $V = U_\sigma U_\rho^{-1} : H \rightarrow K$ . For every  $t \in T$ ,

$$\begin{aligned} V\rho(t) &= U_\sigma \rho_0(t) U_\rho^{-1} \\ &= \sigma(t) U_\sigma U_\rho^{-1} = \sigma(t) V. \end{aligned}$$

Hence every nonzero irreducible representation  $\sigma$  is unitarily equivalent to  $\rho$ .

4. The cited finite-dimensionality theorem applies to a nonzero unital  $C^*$ -algebra with exactly one unitary-equivalence class of nonzero irreducible representations, when that class has a representative on a separable Hilbert space. The representation  $\rho$  on  $H$  satisfies precisely these hypotheses, so  $\dim_{\mathbb{C}} T < \infty$ . But  $\iota$  embeds the infinite-dimensional CAR algebra into  $T$ . This is the contradiction.

## Main citations

- The stated existence or structural result —  
MathlibAnnex.CStarAlgebra.CAR.not\_isIrreducible\_of\_separable
- Unitary equivalence of all nonzero irreducible representations —  
MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint
- Unit preservation forced by nonzero irreducibility —  
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalRepresentation.map\_one\_eq\_one\_of\_isIrreducible
- Unital rebundling with unchanged operators —  
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalRepresentation.toUnital
- A single unitary-equivalence class of nonzero irreducible representations —  
MathlibAnnex.Analysis.CStarAlgebra.Representation.IsSingletonIrreducibleModelAmongNonUnital
- Finite dimensionality when the unique irreducible class has a separable representative —  
MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional\_algebra\_of\_singleton\_amongNonUnital

Supporting route explanation

## Lean source signature (exact)

```
theorem not_isIrreducible_of_separable
  {H : Type v} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
  [CompleteSpace H] [TopologicalSpace.SeparableSpace H]
  (rho : NonUnitalRepresentation (A := AtomicCounterexampleAlgebra) (H := H)) :
  ¬ rho.IsIrreducible
```

| In the source                                                                               | Mathematical meaning                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>AtomicCounterexampleAlgebra</code>                                                    | The one fixed unital shell-generated CAR target $T$ , with its established infinite dimension and unique nonzero irreducible representation class.                                                                                                                                      |
| <code>{H : Type v}; [InnerProductSpace <math>\mathbb{C}</math> H]; [CompleteSpace H]</code> | An arbitrary complete complex Hilbert space $H$ in the independent universe $v$ .                                                                                                                                                                                                       |
| <code>[TopologicalSpace.SeparableSpace H]</code>                                            | The space $H$ has a countable norm-dense subset. The separability condition belongs to $H$ , not to the algebra $T$ .                                                                                                                                                                   |
| <code>rho : NonUnitalRepresentation (A := AtomicCounterexampleAlgebra) (H := H)</code>      | Any complex-linear star representation $\rho : T \rightarrow B(H)$ , without an initial unit-preservation or faithfulness requirement. $T$ itself is still unital.                                                                                                                      |
| <code>¬ rho.IsIrreducible</code>                                                            | This $\rho$ cannot be both nonzero and irreducible. Here irreducibility includes nonzeroness and absence of proper nonzero closed reducing subspaces. In particular, every nonzero such representation is reducible; the conclusion is not a ban on faithful reducible representations. |

Further source notes: `NonUnitalRepresentation` means a  $*$ -representation with no requirement that  $\rho \ 1 = 1$ ; it does not mean that the represented algebra lacks an identity. The local `pi := rho.toUnital hrho` in the linked proof is the same  $\rho$  after  $\rho(1) = I_H$  has been proved, not the direct-sum representation  $\pi$  of CAR. The parameter `v` allows the Hilbert space  $H$  to lie in an independent universe; the comparison spaces in that finite-dimensionality theorem lie in the universe of  $T$ .

Earlier published Card and PDF

#### ▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.not_isIrreducible_of_separable`

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